

**MODEL PARAMETERS AND ITS SIZE FOR TRAINING, USE
AND MODEL FILE**

For Calculation:	NAME	DIMS			TOTAL VALUES	SIZE	UNIT
ATTENTION	<i>KdotQ</i>	4096	4096	1	16777216	64	MB
MATRIX	<i>Key (K)</i>	4096	64	1	262144	1	
	<i>Query (Q)</i>	4096	64	1	262144	1	
	<i>Horizontal (H)</i>	64	4096	1	262144	1	
	<i>Vertical (V)</i>	64	4096	1	262144	1	
MLP	<i>Hor</i>	64	64	16	65536	0.25	
	<i>Ver</i>	64	64	16	65536	0.25	
Continuous Calculation for every head	KEYS	4096	64	1	262144	1	
	QUERIES	4096	64	1	262144	1	
	EV	4096	64	1	262144	1	
	Hlayers	64	16	2	2048	0.0078125	
	Activations	64	16	2	2048	0.0078125	
	gweights (hor)	64	64	16	65536	0.25	
	gweights (ver)	64	64	16	65536	0.25	
CACHE	QK'	64	64	1	4096	0.015625	
	QV'	64	64	1	4096	0.015625	
	KH'	64	64	1	4096	0.015625	
Complete Model Training Parameters and Size (one parallel is number of attention heads working in parallel)				Attention	18890752	72.0625	
				One Parallel:	151126016	576.5	
				Model:	38688260096	144.125	GB
Very Small	TOKENS	1	64	32768	2097152	8	MB
	1D	3	64	2048	393216	1.5	
TOTAL:					38690750464	144.1342773	GB

Data type	float
Bytes	4
Embedding	64
Layers (PAs)	8
Blocks (CAs)	8
Attention Per layer (IAs)	32
Layers of MLP	16
Context	4096
Height	4096
Number of MLP/Head	2
Incomplete Attention IA	Single Head
Partial Attention (Layer)	Consecutive IAs
Block (Complete Attention)	Parallel PAs
Transformer	Multi or Single CA
One Parallel	Heads in parallel ops.

NAME	PARAMETERS	SIZE (MB)	GB
MLP	268435456	1024	1
MAT	2147483648	8192	8
CACHE	25165824	96	0.09375
FOR TRAINING	2415919104	9216	9
FOR USE	293601280	1120	1.09375
TOTAL	2441084928	9312	9.09375

For Calculation:	NAME	DIMS			TOTAL VALUES	SIZE	UNIT
ATTENTION	<i>KdotQ</i>	4096	4096	1	16777216	64	MB
CACHE	QK'	64	64	1	4096	0.015625	
	QV'	64	64	1	4096	0.015625	
	KH'	64	64	1	4096	0.015625	
MLP	Hor	64	64	16	65536	0.25	
	Ver	64	64	16	65536	0.25	
	Hlayers	64	16	2	2048	0.0078125	
	Activations	64	16	2	2048	0.0078125	
For Model Use Parameters and Size				Attention:	16924672	64.5625	
				One Parallel:	135397376	516.5	
				Model:	4332716032	16.140625	GB
Very Small	TOKENS	1	64	32768	2097152	8	MB
	EV	1	64	8388608	536870912	2048	
	1D	3	64	256	49152	0.1875	
TOTAL:					4871733248	18.14862061	GB

For Trained Params:	NAME	DIMS			Attention	Model	SIZE (MB)
MATRIX	<i>Key (K)</i>	4096	64	1	262144	536870912	2048
	<i>Query (Q)</i>	4096	64	1	262144	536870912	2048
	<i>Horizontal (H)</i>	64	4096	1	262144	536870912	2048
	<i>Vertical (V)</i>	64	4096	1	262144	536870912	2048
MLP	Hor	64	64	16	65536	134217728	512
	Ver	64	64	16	65536	134217728	512
CACHE	QK'	64	64	1	4096	8388608	32
	QV'	64	64	1	4096	8388608	32
	KH'	64	64	1	4096	8388608	32
MODEL PARAMETERS					1191936	2441084928	9312